

LASK, Poland

WMO: 124540

Lat: 51.552N

Lon: 19.179E

Elev: 193

StdP: 99.03

Time Zone: 1.00 (EUC)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.8	-12.1	-17.3	0.8	-13.6	-15.0	1.0	-11.2	11.6	4.4	10.7	3.7	2.4	100	0.553

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	30.7	19.7	28.9	19.1	27.1	18.4	21.1	27.9	20.3	26.6	19.5	25.1	3.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	14.2	23.0	18.2	13.4	22.3	17.4	12.8	21.7	61.8	28.1	59.1	26.7	56.2	25.1	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.6	8.3	7.2	DB	-20.0	34.5	4.1	1.3	-23.0	35.4	-25.4	36.2	-27.7	36.9	-30.7	37.8	(4)
(5)			WB	-19.3	23.1	3.9	0.8	-22.1	23.7	-24.4	24.2	-26.6	24.6	-29.4	25.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.0	-2.4	-0.9	3.4	9.4	13.8	17.5	19.6	19.1	14.1	8.6	5.0	0.5
	DBStd	8.62	5.27	4.98	4.70	4.29	3.95	3.54	3.53	3.11	3.54	4.08	3.82	4.77
	HDD10.0	1498	383	305	207	62	11	0	0	0	5	77	152	296
	HDD18.3	3562	642	538	462	270	149	58	28	29	133	301	399	554
	CDD10.0	1144	0	0	3	43	128	224	297	282	128	34	4	0
	CDD18.3	167	0	0	0	1	8	32	67	53	7	0	0	0
	CDH23.3	1611	0	0	0	17	98	325	623	467	82	0	0	0
	CDH26.7	468	0	0	0	2	17	84	212	132	21	0	0	0
(14) Wind	WSAvg	3.4	3.7	3.5	3.7	3.4	3.2	3.1	3.0	2.9	3.1	3.3	3.6	3.9
(15) Precipitation	PrecAvg	570	34	32	38	35	61	63	84	54	48	39	42	39
	PrecMax	750	73	72	79	84	158	149	238	122	112	127	105	91
	PrecMin	390	5	4	15	1	19	11	25	20	15	6	0	8
	PrecStd	91	16	16	15	19	30	38	51	25	24	27	20	20
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.7	11.4	17.7	25.8	28.8	32.2	33.5	32.6	29.4	21.6	15.7	11.6
		MCWB	7.7	8.1	10.5	15.3	18.0	20.8	20.1	20.0	18.5	15.5	11.9	9.2
	2%	DB	7.5	9.3	14.9	22.1	26.0	29.0	31.3	29.7	25.5	19.4	13.5	9.1
		MCWB	6.0	6.8	8.8	13.4	16.9	18.9	19.5	19.5	17.0	14.5	10.9	7.8
	5%	DB	5.9	7.2	12.7	19.7	23.7	26.9	29.1	28.0	23.0	17.0	11.6	7.7
		MCWB	4.5	5.0	8.0	11.8	15.8	18.1	19.6	18.9	16.1	13.3	9.8	6.5
	10%	DB	4.3	5.3	10.3	17.3	21.3	25.0	26.9	26.1	20.8	14.7	10.2	6.3
		MCWB	3.1	3.8	6.9	10.7	14.4	16.8	18.8	18.2	15.2	12.2	8.8	5.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.2	9.1	11.5	15.7	19.2	21.8	22.2	22.0	19.5	16.6	12.5	9.2
		MCDB	9.4	10.4	16.3	23.9	26.0	28.9	28.3	28.6	27.6	20.3	14.7	10.7
	2%	WB	6.2	6.9	9.8	14.1	17.8	20.3	21.2	20.7	18.0	15.0	11.2	8.0
		MCDB	7.4	9.1	13.8	20.7	23.9	26.9	27.9	27.3	22.9	18.3	13.0	9.1
	5%	WB	4.7	5.2	8.5	12.7	16.5	19.1	20.5	19.9	16.9	13.7	10.0	6.7
		MCDB	5.7	6.8	11.9	18.1	22.3	24.7	26.7	26.2	21.6	16.4	11.4	7.4
	10%	WB	3.2	3.9	7.1	11.3	15.3	18.0	19.7	19.0	15.7	12.5	8.9	5.4
		MCDB	4.2	5.3	10.0	16.1	20.2	23.1	25.1	24.2	19.9	14.6	10.1	6.2
(35) Mean Daily Temperature Range	5% DB	MDBR	4.8	6.1	8.0	10.4	10.5	11.0	10.9	11.0	9.8	7.7	5.5	4.7
		MCDBR	5.1	8.0	11.9	14.5	14.0	14.2	14.1	14.1	13.6	10.9	8.0	5.3
		MCWBR	4.3	5.8	7.1	7.5	6.6	6.0	5.0	5.6	6.3	6.6	5.7	4.2
	5% WB	MCDBR	5.1	7.4	10.7	12.8	12.5	12.3	12.2	12.2	11.9	10.0	7.6	5.0
		MCWBR	4.5	5.7	6.8	7.1	6.4	6.0	5.1	5.4	6.1	6.4	5.6	4.3
(40) Clear-Sky Solar Irradiance	taub		0.316	0.342	0.379	0.430	0.423	0.418	0.447	0.434	0.399	0.370	0.348	0.313
	taud		2.268	2.266	2.219	2.132	2.207	2.252	2.186	2.223	2.291	2.341	2.327	2.309
	Ebn at Noon		677	766	805	804	832	839	804	793	778	726	634	619
	Edh at Noon		68	89	113	139	135	130	137	126	105	83	65	56
(44) All-Sky Solar Radiation	RadAvg		0.74	1.44	2.65	4.15	5.17	5.41	5.06	4.53	3.25	1.83	0.84	0.56
	RadStd		0.14	0.26	0.32	0.55	0.54	0.52	0.60	0.38	0.45	0.28	0.11	0.07

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)		+0.51	N/A	N/A	N/A	N/A	N/A	N/A	-164	+78	+30

Nomenclature: See separate page